

Manas Sharma

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EDUCATION

Manipal Institute of Technology

Bachelor of Technology, Computer Science – 8.5 cgpa

Bengaluru

Jul 2023 – Jul 2027 (Expected)

EXPERIENCE

Backend Engineering Intern

NRI Banking & Investment Infrastructure

RupeeFlo (Fintech)

May 2026 – Jul 2026

- Migrated document-generation pipelines from **Supabase** to the core **Django** backend, collapsing a 3-call chain into one API and cutting processing time by **80%**; proved **0% divergence** with a PDF-diff **parity harness**.
- Designed a **3-tier OTP fallback** (WhatsApp → SMS → voice) using **Django, Celery, and Redis** with atomic **SETNX** concurrency locks, rate limits, and idempotent escalation; serving **15000+** users, fixed **27** edge cases.
- Shipped an **AI support draft generator** to production, aggregating **Freshdesk** history and Customer-360 data into structured **LLM tool-use** calls; cut reply drafting time **~90%** with **70%** coverage on 78 real tickets.

Machine Learning Intern

Video Analytics Division

Reliance Industries Limited

May 2025 - Jul 2025

- Built a real-time occupancy detection system using **YOLOv11** and **DeepSort** to track chair usage across **RTSP streams** and webcam feeds, achieving **30+ FPS** throughput with sub-second latency on GPU deployment.
- Designed a **Multi-Camera Manager** with spatial deduplication and priority-based conflict resolution to eliminate duplicate counts across overlapping FOVs, enabling unified multi-feed analytics.

PROJECTS

Distributed Message Broker | Go, Protobuf, gRPC

- Built a distributed message broker on **HashiCorp Raft** consensus and **partition sharding**; measured **262K msg/sec** produce with fsync on (p99 **56ms**), **723K** fsync-off, and **490K msg/sec** consume using a reproducible **gRPC load-test harness**.
- Implemented **leader-follower replication** with ISR-acknowledged produce and transparent **leader-forwarding**; automatic controller and partition-leader **failover** (**~10s**) across **3-node clusters**.
- Developed segmented commit logs with **CRC32C checksums**, and fsync amortized across **500-record batches**; built **Prometheus/Grafana** monitoring and **Docker/Kubernetes** StatefulSet deployments.

VLM Evaluation Pipeline for Driving Scenes | Python, VLMs (Gemini / Groq), Pydantic, Docker

- Built a **VLM evaluation pipeline** comparing **8 prompt strategies** on 100 BDD100K scenes (**800 evaluations**) across **9 metrics** (hallucination, miss rate, count MAE, image-grounded **LLM-as-judge**); **schema-constrained prompting** cut miss rate **28.9% to 26.2%** and count error **5%** ($p < 0.05$, paired permutation tests).
- Developed an **error-analysis agent** that mines results for failure patterns and **auto-generates improved prompts**; **checkpoint/resume** orchestration recovered the full 800-call run across daily API quota limits.

Marketplace Price Prediction | PyTorch, FastAPI, Next.js, MongoDB, Docker

- Trained a **BiLSTM + attention** model on **1.48M Mercari listings**, reaching **0.420 RMSLE**; beat XGBoost/LightGBM baselines and matched a fine-tuned **DistilBERT** with **4.5× fewer parameters**.
- Built a **FastAPI** serving layer with rate limiting, caching, **SHAP** explanations, and an **LLM listing-critique endpoint**; deployed on **GCP Cloud Run** with GitHub Actions CI.

TECHNICAL SKILLS

Languages: Go, Python, SQL, TypeScript, C++, Java

Frameworks & Libraries: gRPC, Protobuf, Django/DRF, FastAPI, Celery, Next.js, React.js, Node.js, LangChain

Machine Learning: PyTorch, LLMs, RAG, Scikit-learn, OpenCV, ONNX

Databases & Infrastructure: PostgreSQL, MongoDB, Redis, Docker, Kubernetes, GCP, AWS, Linux

CS Fundamentals: DSA, OOP, Multithreading & Concurrency, Operating Systems, DBMS, Computer Networks

ACHIEVEMENTS & CERTIFICATIONS

- Solved **300+** problems on LeetCode and CodeForces. Mentored **100+** students in technical workshops
- Led my team to win Atos Hackathon 2026** (100+ teams) and **4th place, CCL Hackathon 2026** (500+ teams).
- Certifications:** Deep Learning Specialization, Azure AI Fundamentals.